Граници на редици с неопределеност от вида $[1^\infty]$

Когато имаме неопределеност от вида $[1^\infty]$, т.е. редица $c_n^{b_n}$, където $c_n \xrightarrow[n\to\infty]{} 1$ и $b_n \xrightarrow[n\to\infty]{} \infty$, свеждаме общия член до $\left(1+\frac{a_n}{b_n}\right)^{b_n}$ и използваме границата $\lim\limits_{n\to\infty}\left(1+\frac{a_n}{b_n}\right)^{b_n} = e^{\lim_{n\to\infty} a_n}$.

Задачи

1) $\lim\limits_{n\to\infty}\left(1+\frac{1}{n-1}\right)^n$ 2) $\lim\limits_{n\to\infty}\left(1+\frac{2}{n}\right)^n$ 3) $\lim\limits_{n\to\infty}\left(1+\frac{1}{n}\right)^{3n}$ 4) $\lim\limits_{n\to\infty}\left(1-\frac{1}{n}\right)^n$

5) $\lim\limits_{n\to\infty}\left(1+\frac{1}{n^2}\right)^n$ 6) $\lim\limits_{n\to\infty}\left(1-\frac{1}{n^2}\right)^n$ 7) $\lim\limits_{n\to\infty}\left(1+\frac{1}{n}\right)^{n^2}$ 8) $\lim\limits_{n\to\infty}\left(1+\frac{n}{3n^2-1}\right)^{n^2+2}$

9) $\lim\limits_{n\to\infty}\left(\frac{n^2+4n+3}{n^2+5n+6}\right)^n$ 10) $\lim\limits_{n\to\infty}\left(\frac{n^3+n+1}{n^3+2n+5}\right)^n$ 11) $\lim\limits_{n\to\infty}\left(\frac{2n^2+n+2}{2n^2+3}\right)^{\frac{n^2}{n+1}+\sin n}$

12) $\lim\limits_{n\to\infty}\left(\frac{n^2+1}{n^2+n}\right)^{n\sqrt{n}}$ 13) $\lim\limits_{n\to\infty}\left(\frac{n^2+n}{n^2-2}\right)^{n\sqrt{n}}$.

Решения

Навсякъде с червено е означено a_n, а със синьо: b_n.

1) $\lim\limits_{n\to\infty}\left(1+\frac{1}{n-1}\right)^n = \lim\limits_{n\to\infty}\left(1+\frac{n}{n(n-1)}\right)^n = e^{\lim_{n\to\infty}\frac{n}{n-1}} = e^1 = e.$

Забележка. Задачата може да се реши и чрез записване на степента n като $(n-1)+1$ и се използва произведение на граници.

2) $\lim\limits_{n\to\infty}\left(1+\frac{2}{n}\right)^n = e^{\lim_{n\to\infty} 2} = e^2.$

3) $\lim\limits_{n\to\infty}\left(1+\frac{1}{n}\right)^{3n} = \lim\limits_{n\to\infty}\left(1+\frac{3}{3n}\right)^{3n} = e^{\lim_{n\to\infty} 3} = e^3.$

4) $\lim\limits_{n\to\infty}\left(1-\frac{1}{n}\right)^n = \lim\limits_{n\to\infty}\left(1+\frac{-1}{n}\right)^n = e^{\lim_{n\to\infty} -1} = e^{-1} = \frac{1}{e}.$

5) $\lim\limits_{n\to\infty}\left(1+\frac{1}{n^2}\right)^n = \lim\limits_{n\to\infty}\left(1+\frac{1}{n.n}\right)^n = e^{\lim_{n\to\infty}\frac{1}{n}} = e^0 = 1.$

6) $\lim\limits_{n\to\infty}\left(1-\frac{1}{n^2}\right)^n = \lim\limits_{n\to\infty}\left(1+\frac{-1}{n.n}\right)^n = e^{\lim_{n\to\infty}-\frac{1}{n}} = e^0 = 1.$

7) $\lim\limits_{n\to\infty}\left(1+\frac{1}{n}\right)^{n^2} = \lim\limits_{n\to\infty}\left(1+\frac{n}{n^2}\right)^{n^2} = e^{\lim_{n\to\infty} n} = [e^\infty] = \infty.$

8) $\lim\limits_{n\to\infty}\left(1+\dfrac{n}{3n^2-1}\right)^{n^2+2}=\lim\limits_{n\to\infty}\left(1+\dfrac{n(n^2+2)}{(3n^2-1)(n^2+2)}\right)^{n^2+2}=e^{\lim_{n\to\infty}\frac{n^3+2n}{3n^2-1}}=[e^{\infty}]=\infty.$

9)

$$\lim_{n\to\infty}\left(\frac{n^2+4n+3}{n^2+5n+6}\right)^n=\lim_{n\to\infty}\left(\frac{n^2+4n+3+n-n+3-3}{n^2+5n+6}\right)^n=\lim_{n\to\infty}\left(1+\frac{-n-3}{n^2+5n+6}\right)^n$$
$$=\lim_{n\to\infty}\left(1+\frac{n(-n-3)}{n(n^2+5n+6)}\right)^n=e^{\lim_{n\to\infty}\frac{-n^2-3n}{n^2+5n+6}}=e^{-1}=\frac{1}{e}.$$

Забележка. Ако се съобрази, че $n^2+4n+3=(n+3)(n+1)$ и $n^2+5n+6=(n+3)(n+2)$, то общият член е $\left(\dfrac{n+1}{n+2}\right)^n$ и задачата се довършва с добавяне и изваждане на 1 в числителя, разделяне на дробта и умножаване и разделяне на n.

10)

$$\lim_{n\to\infty}\left(\frac{n^3+n+1}{n^3+2n+5}\right)^n=\lim_{n\to\infty}\left(\frac{n^3+n+1+n-n+4-4}{n^3+2n+5}\right)^n=\lim_{n\to\infty}\left(1+\frac{-n-4}{n^3+2n+5}\right)^n$$
$$=\lim_{n\to\infty}\left(1+\frac{n(-n-4)}{n(n^3+2n+5)}\right)^n=e^{\lim_{n\to\infty}\frac{-n^2-4n}{n^3+2n+5}}=e^{\lim_{n\to\infty}-\frac{1}{n}}=e^0=1.$$

11)

$$\lim_{n\to\infty}\left(\frac{2n^2+n+2}{2n^2+3}\right)^{\frac{n^2}{n+1}+\sin n}=\lim_{n\to\infty}\left(1+\frac{n-1}{2n^2+3}\right)^{\frac{n^2}{n+1}+\sin n}$$
$$=\lim_{n\to\infty}\left(1+\frac{\left(\frac{n^2}{n+1}+\sin n\right)(n-1)}{\left(\frac{n^2}{n+1}+\sin n\right)(2n^2+3)}\right)^{\frac{n^2}{n+1}+\sin n}$$
$$=e^{\lim_{n\to\infty}\frac{\left(\frac{n^2}{n+1}+\sin n\right)(n-1)}{2n^2+3}}=e^{\frac{1}{2}}=\sqrt{e},$$

защото $\lim\limits_{n\to\infty}\dfrac{\frac{n^2}{n+1}+\sin n}{n}=\lim\limits_{n\to\infty}\dfrac{n^2}{n^2+n}+\lim\limits_{n\to\infty}\dfrac{\sin n}{n}=1+0=1$, т.е. $\frac{n^2}{n+1}+\sin n\sim n$, като $n-1\sim n$ и $2n^2+3\sim 2n^2$, откъдето

$$\lim_{n\to\infty}\frac{\left(\frac{n^2}{n+1}+\sin n\right)(n-1)}{2n^2+3}=\lim_{n\to\infty}\frac{n.n}{2n^2}=\frac{1}{2}.$$

12)

$$\lim_{n\to\infty}\left(\frac{n^2+1}{n^2+n}\right)^{n\sqrt{n}}=\lim_{n\to\infty}\left(1+\frac{-n+1}{n^2+n}\right)^{n\sqrt{n}}=\lim_{n\to\infty}\left(1+\frac{n\sqrt{n}(-n+1)}{n\sqrt{n}(n^2+n)}\right)^{n\sqrt{n}}$$
$$=e^{\lim_{n\to\infty}\frac{n\sqrt{n}(-n+1)}{n^2+n}}=[e^{-\infty}]=0,$$

понеже $n\sqrt{n}(-n+1)\sim -n^2\sqrt{n}$ и $n^2+n\sim n^2$, откъдето

$$\lim_{n\to\infty}\frac{n\sqrt{n}(-n+1)}{n^2+n}=\lim_{n\to\infty}\frac{-n^2\sqrt{n}}{n^2}=\lim_{n\to\infty}-\sqrt{n}=-\infty.$$

13)

$$\lim_{n\to\infty}\left(\frac{n^2+n}{n^2-2}\right)^{n\sqrt{n}}=\lim_{n\to\infty}\left(1+\frac{n+2}{n^2-2}\right)^{n\sqrt{n}}=\lim_{n\to\infty}\left(1+\frac{n\sqrt{n}(n+2)}{n\sqrt{n}(n^2-2)}\right)^{n\sqrt{n}}$$
$$=e^{\lim_{n\to\infty}\frac{n\sqrt{n}(n+2)}{n^2-2}}=[e^{\infty}]=\infty.$$